

Information Theory

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Outline

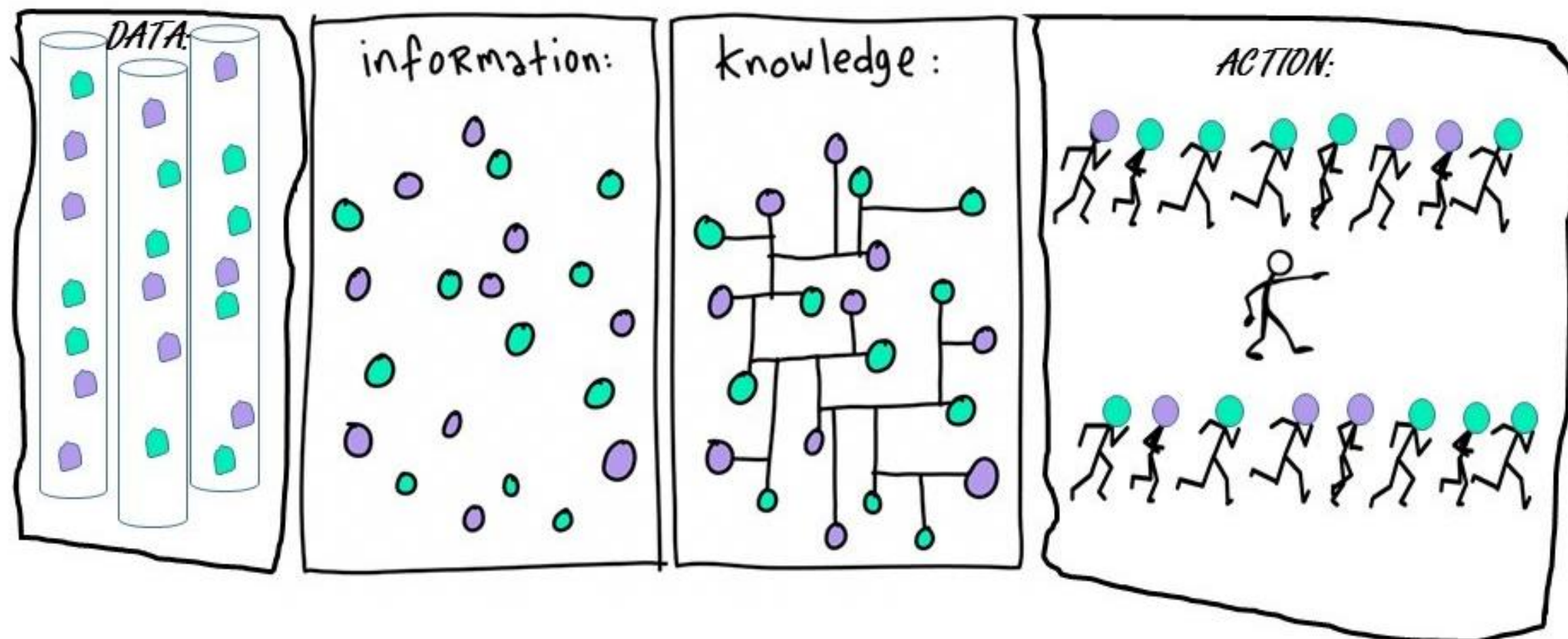
- Motivation 
- Entropy
- Conditional Entropy and Mutual Information
- Cross-Entropy and KL-Divergence

Uncertainty and Information

Information is processed data whereas **knowledge** is **information** that is modeled to be useful.

You need **information** to be able to get **knowledge**

- **More information when an unlikely event occurs than when something certain occurs** (in fact, it should be zero when the event is certain)
- **Example:** You are in sunny Los Angeles, California and you are told it did not rain yesterday → **not a lot of information since it rarely rains in SoCal**

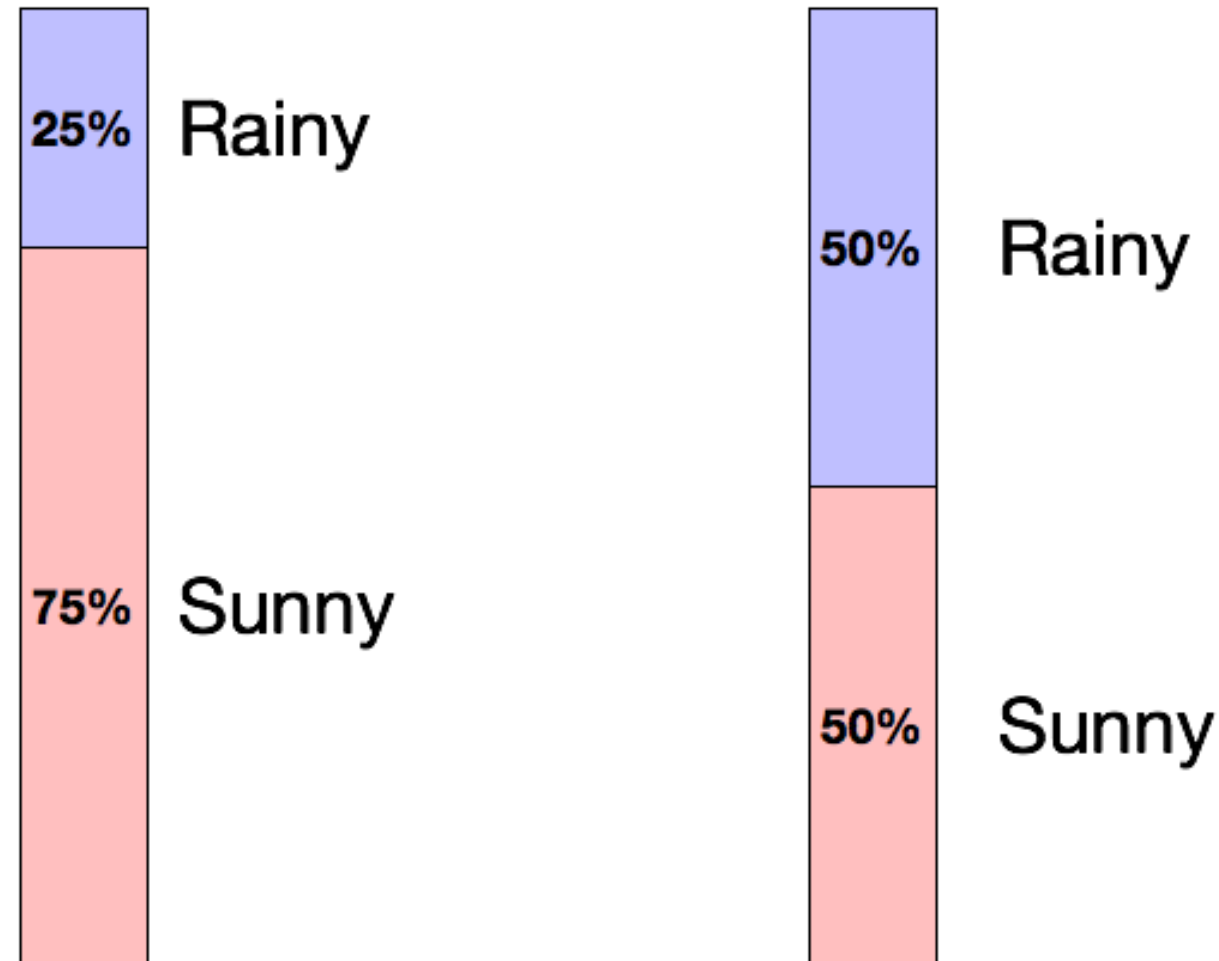


Created by Bruce Campbell: “DIKA – ancient Chinese saying for get up and DO! Data-Information-Knowledge-Action.”

Information Theory

- Information theory is a mathematical framework which addresses questions like:
 - How much information does a random variable carry about?
 - How efficient is a hypothetical code, given the statistics of the random variable?
 - How can we encode more efficiently?
 - Is the **information carried by different random variables complementary or redundant?**

Uncertainty and Information



Which day is more uncertain?

How do we quantify uncertainty?

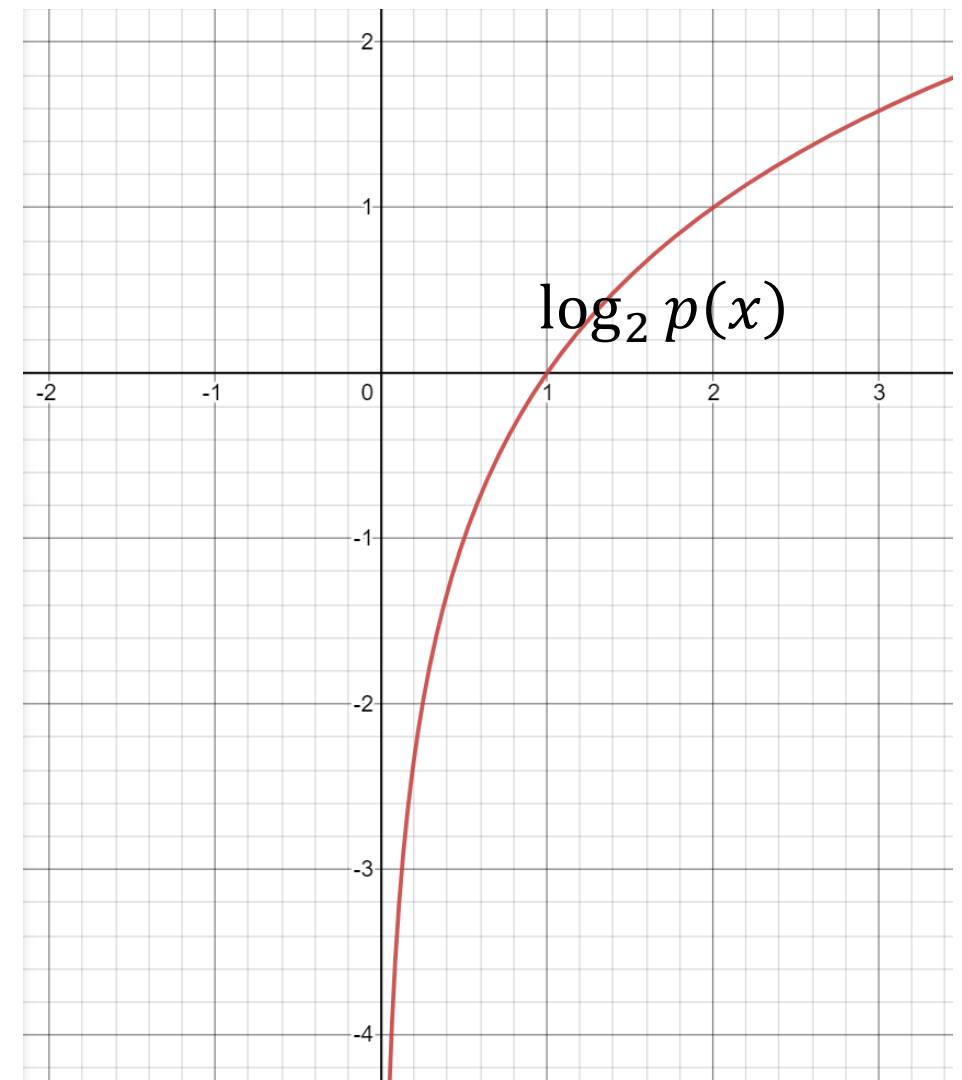
Less likely → More uncertainty → More information →
High entropy

Information

We can associate our measure of information with probability of an event occurring. Let X be a random variable with distribution $p(x)$:

$$I(x) = \log_2 \frac{1}{p(x)} = -\log_2 p(x)$$

High probability means less information



Information: Example

$X = [\text{cat}, \quad]$

Example: is a picture worth 1,000 words?

- Information obtained by a random word from a 100,000 word vocabulary:

$$I(word) = \log_2 \left(\frac{1}{p(x)} \right) = \log_2 \left(\frac{1}{1/100,000} \right) = 16.61 \text{ bits}$$

- A 1,000-word document from same source:

$$I(document) = 1000 \times I(word) = 16,610 \text{ bits}$$

- A 640 x 480 pixel, 16-greyscale picture (each pixel has 16 bits information):

$$I(picture) = \log_2 \left(\frac{1}{1/16^{640 \times 480}} \right) = 1,228,800 \text{ bits}$$

A picture is worth (a lot more than) 1,000 words!

Motivation: Data compression

- Suppose we observe a sequence of events
 - Coin tosses
 - Words in a language
 - Notes in a song
 - etc.
- We want to record the sequence of events in the smallest possible space
- In other words, we want the shortest representation which preserves the information
- Another way to think about this: **how much information does the sequence of events actually contain?**

Example: Data compression

- Consider the problem of recording coin tosses in unary

T, T, T, T, H

- Approach 1:**

Heads	Tails
0	00

00, 00, 00, 00, 0

We used **9** characters

- Which one has a higher probability: T or H?
- Which one should carry more information: T or H?

Example: Data compression

- Consider the problem of recording coin tosses in unary

T, T, T, T, H

- Approach 2:**

Heads	Tails
00	0

0, 0, 0, 0, 00

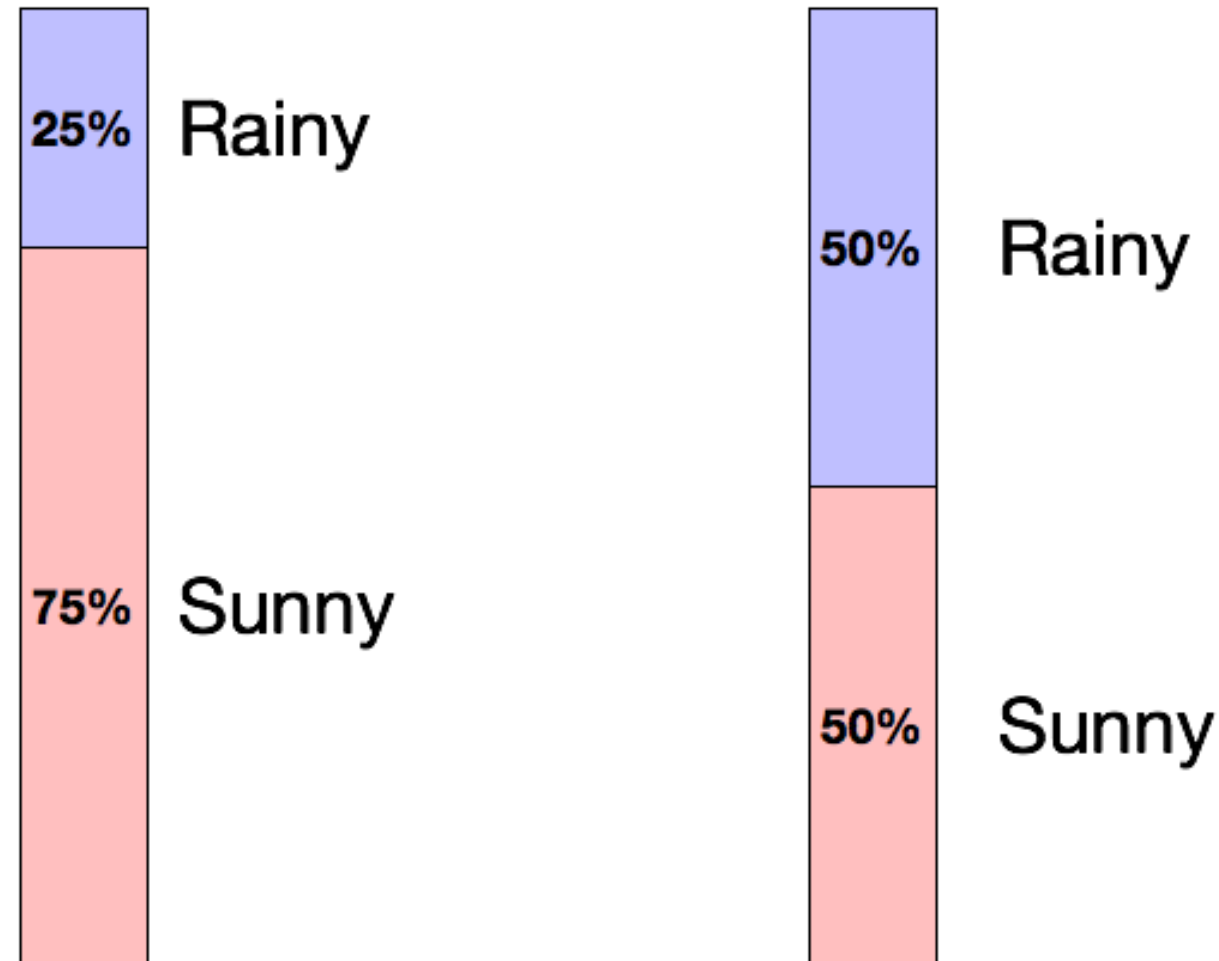
We used **6** characters

- Which one has a higher probability: T or H?
- Which one should carry more information: T or H?

Motivation: Data compression

- Frequently occurring events should have short encodings
- We see this in English with words such as “a”, “the”, “and”, etc.
- We want to maximize the information-per-character
- Seeing common events provides little information
- Seeing uncommon events provides a lot of information

Uncertainty and Information



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Recap

- Prob vs. Likelihood
- MLE
- Information

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Entropy

- Average amount of information to encode a random variable X with respect to its distribution $p(x)$ is the entropy $H(x)$:

$$E[g(x)] = \sum_x g(x)p(x)$$

$$H(x) = E[I(x)] = \sum_x I(x)p(x) = \sum_x p(x) \log_2 \left(\frac{1}{p(x)} \right) = - \sum_x p(x) \log_2 p(x)$$

Considering a random variable X with k possible states:

$$H(x) = - \sum_{k=1}^K p(x = k) \log_2 p(x = k) = \sum_{k=1}^K p(x = k) \log_2 \frac{1}{p(x = k)}$$

- Entropy is the **average amount of surprise (information)** you gain when observing outcomes of X .

Entropy

- Max possible entropy a random variable X with k possible states is $\log_2 k$
- Entropy is non-negative
- Most efficient code assigns $-\log_2 P(x = k)$ bits to encode the message $x = k$.
 - because it matches the information content of that event.

Example: entropy computation for coin toss

$$H(S) \equiv -(p_+ \log_2 p_+ + p_- \log_2 p_-)$$

Head	0
Tail	6

$$p(H) = \frac{0}{6} = 0, \quad p(T) = \frac{6}{6} = 1$$
$$H = -0 \log_2 0 - 1 \log_2 1 = 0$$

Head	1
Tail	5

$$p(H) = \frac{1}{6}, \quad p(T) = \frac{5}{6}$$
$$H = -\frac{1}{6} \log_2 \frac{1}{6} - \frac{5}{6} \log_2 \frac{5}{6} = 0.65$$

Head	2
Tail	4

$$p(H) = \frac{2}{6}, \quad p(T) = \frac{4}{6}$$
$$H = -\frac{2}{6} \log_2 \frac{2}{6} - \frac{4}{6} \log_2 \frac{4}{6} = 0.92$$

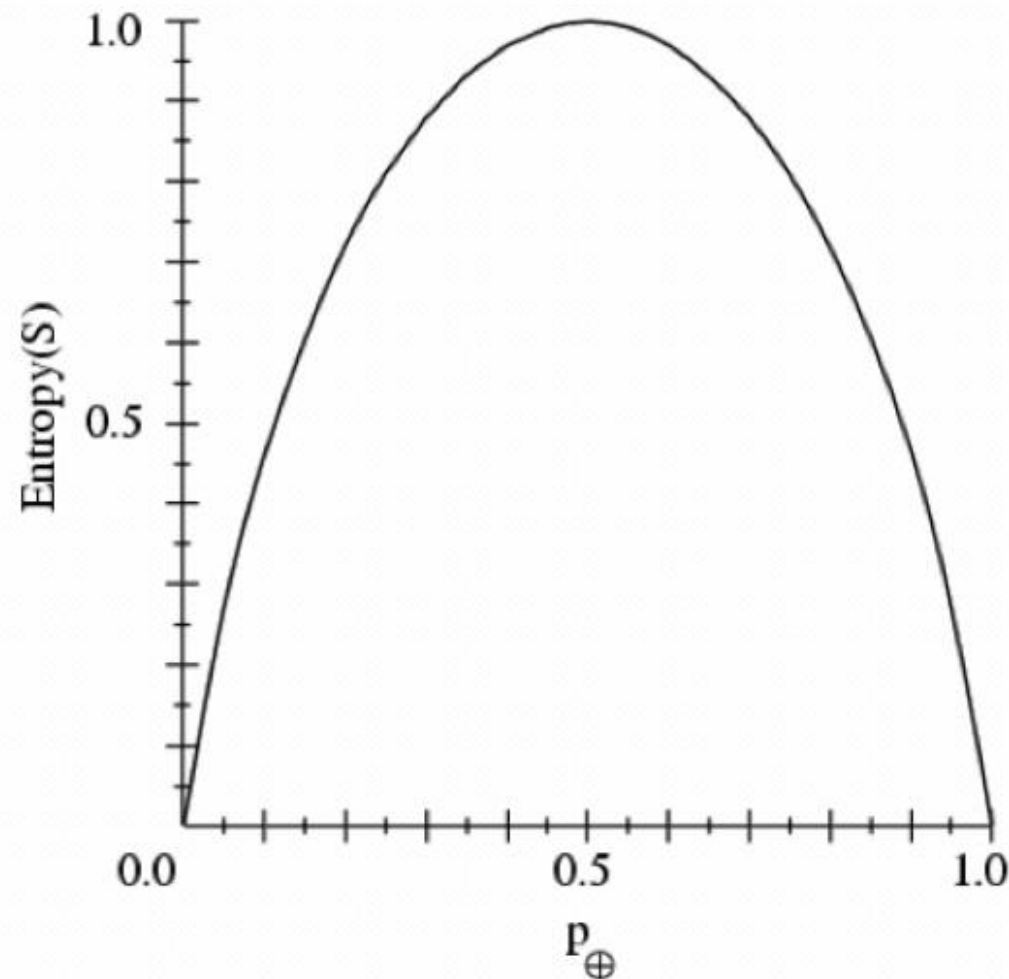
Head	3
Tail	3

$$p(H) = \frac{1}{2}, \quad p(T) = \frac{1}{2}$$
$$H = -\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2} = 1$$

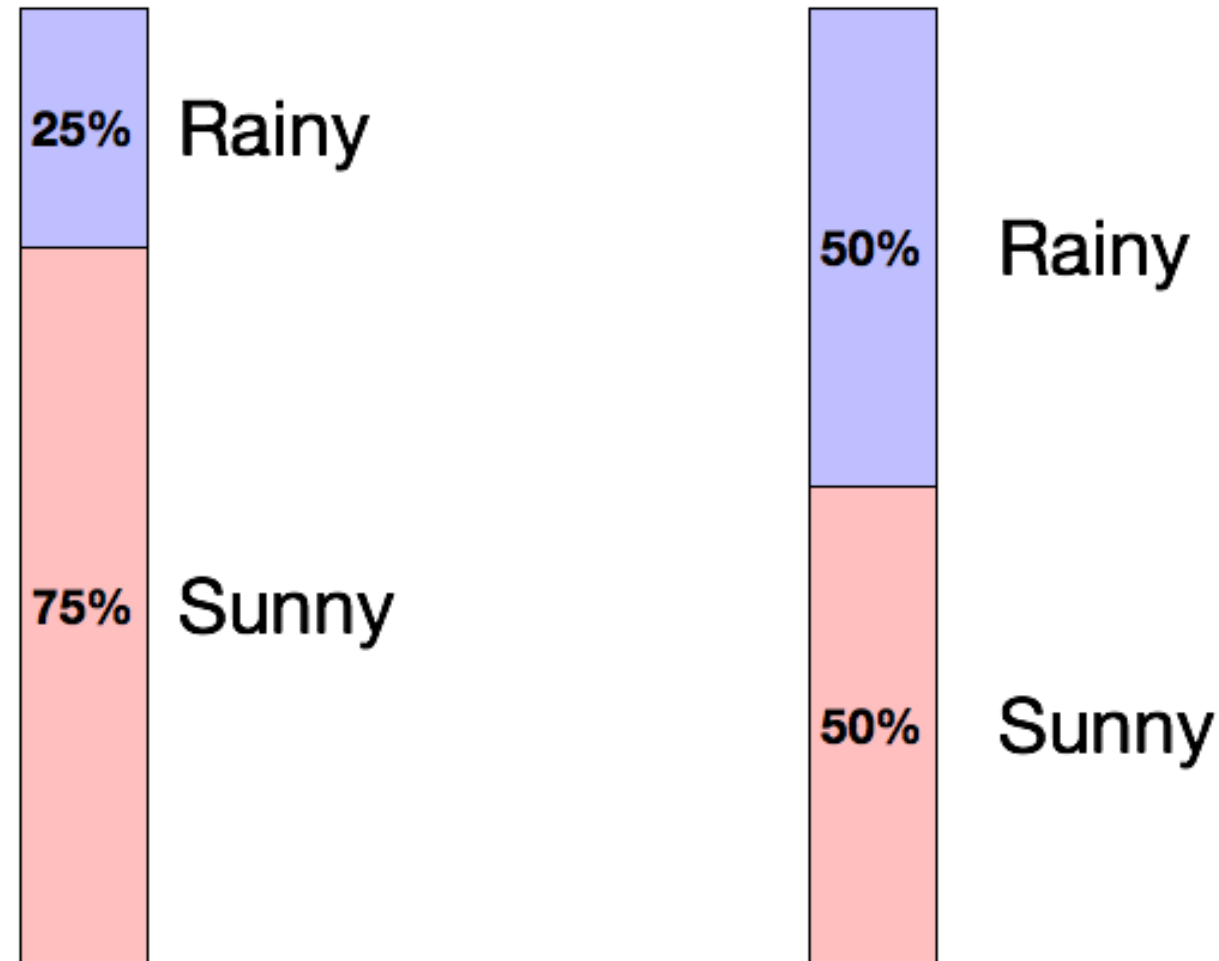
Example: entropy

- S is a sample of coin flips
- p_+ is the proportion of heads in S
- p_- is the proportion of tails in S
- Entropy measures the uncertainty of S

$$H(S) \equiv -(p_+ \log_2 p_+ + p_- \log_2 p_-)$$



Uncertainty and Information



Which day is more uncertain?

How do we quantify uncertainty?

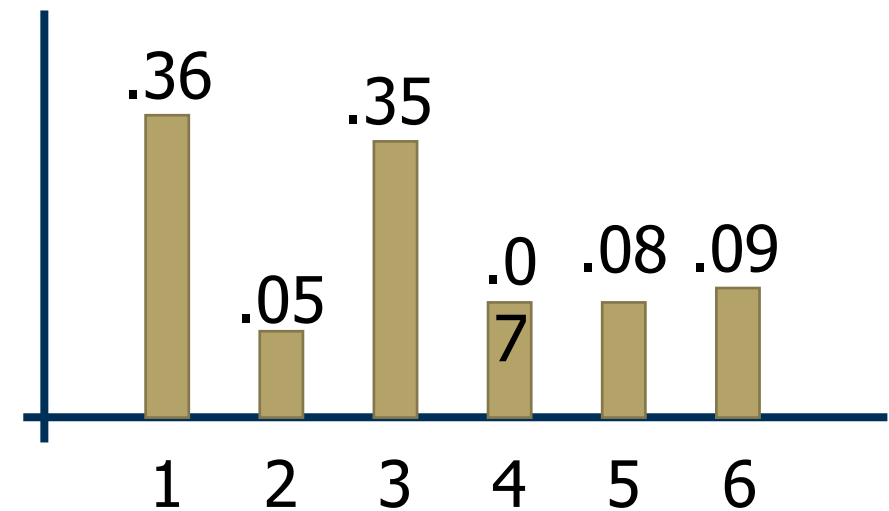
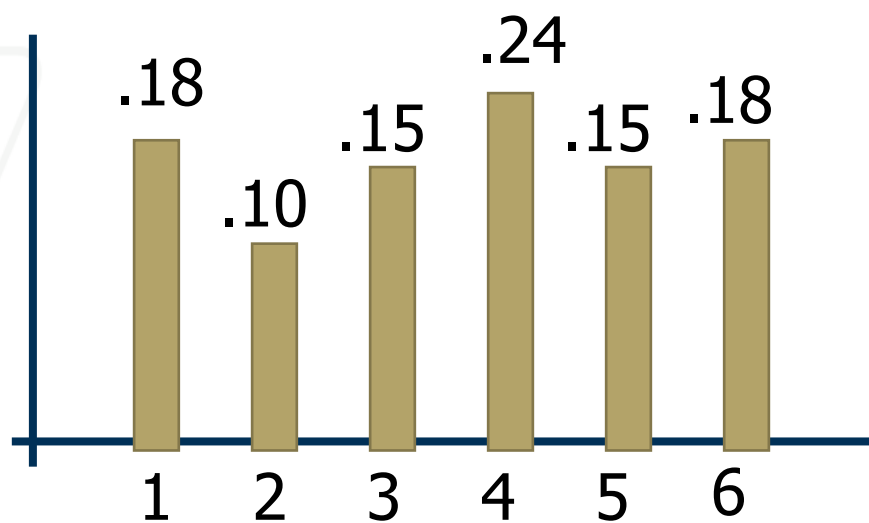
Less likely → More uncertainty → More information →
High entropy

Properties of Entropy

- Non-negative: $H(P) \geq 0$
- Invariant with respect to permutation of its inputs:
$$H(p_1, p_2, \dots, p_k) = H(p_{\tau(1)}, p_{\tau(2)}, \dots, p_{\tau(k)})$$
- For any other probability distribution $\{q_1, q_2, \dots, q_k\}$
$$H(P) = \sum_i p_i \log \frac{1}{p_i} < \sum_i p_i \log \frac{1}{q_i}$$
- $H(P) \leq \log_2 k$, with equality iff $p_i = \frac{1}{k}, \forall i$
- The further P is from uniform, the lower the entropy

Example: entropy for rolling a die

$$H(x) = - \sum_{k=1}^K p(x = k) \log_2 p(x = k)$$



In context: Why does Entropy matter?

Entropy measures uncertainty.

- It tells us how much surprise to expect on average.
- High entropy = we should expect to learn more once the outcome is revealed.

Entropy helps us compare systems

- Coin that always lands heads $\rightarrow$ entropy = 0 (no uncertainty, no info).
- Fair coin $\rightarrow$ entropy = 1 bit (max for binary outcomes).
- Six-sided die $\rightarrow$ entropy $\approx$ 2.58 bits (more uncertainty than a coin).

👉 Why helpful?

- It lets us compare “information richness” between random processes.
- Used in feature selection. Features with higher entropy have more capacity to carry information.

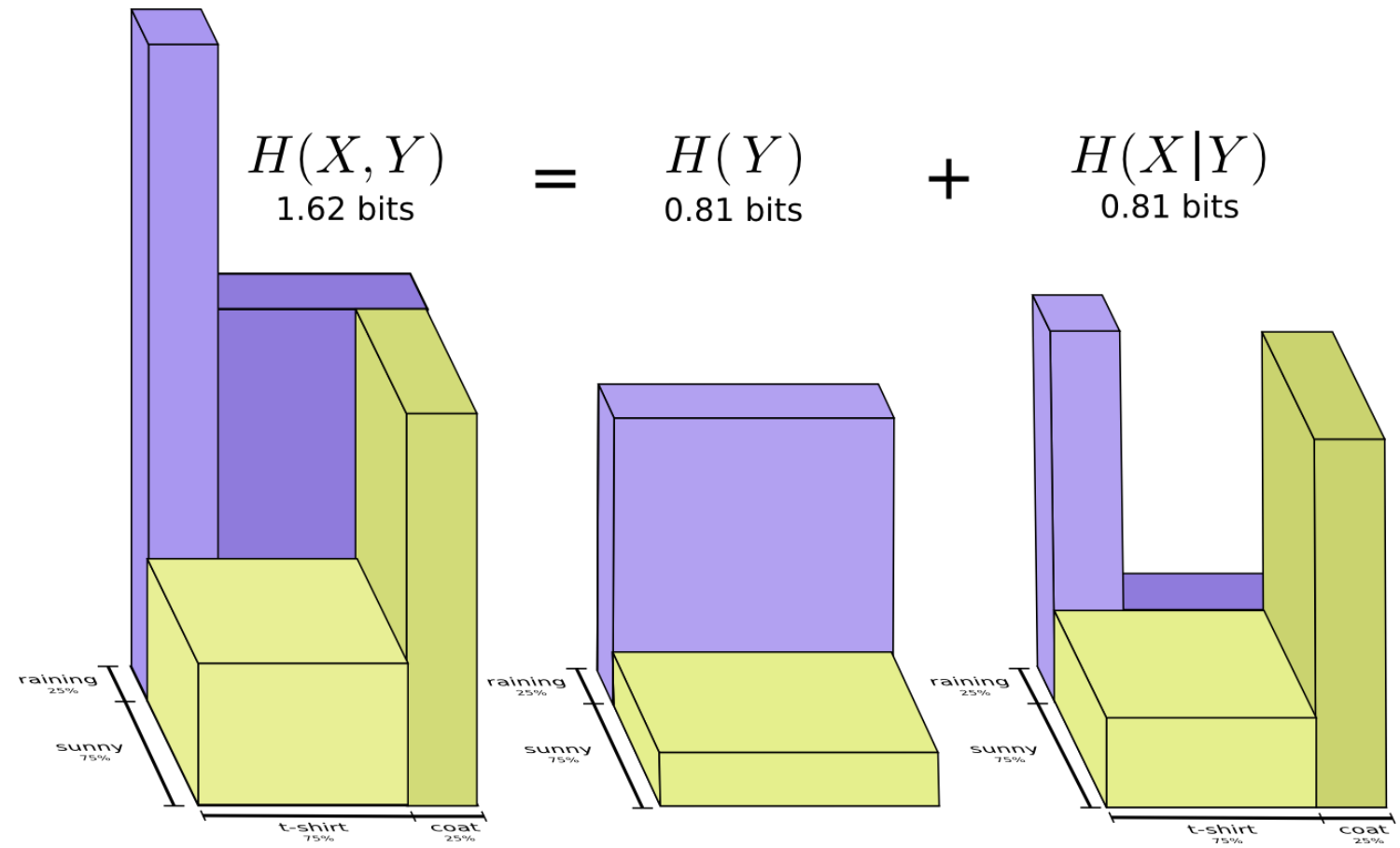
Big picture intuition

- Entropy matters because it tells us how much “space” we need to represent uncertainty.
- The more uncertain the world is, the more information we gain by observing it — and entropy quantifies that.

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Entropy



Joint Entropy

- The joint entropy measures the entropy of a joint probability distribution of two random variables X and Y

$$H(X, Y) = \sum_{x \in X, y \in Y} p(x_i, y_i) \log \frac{1}{p(x_i, y_i)}$$

- It helps us quantify the **dependence** between different variables and how much information they share
- From the product rule we can also see that

$$H(X, Y) = H(X) + H(Y|X)$$

Joint Entropy

humidity	Temperature			
	cold	mild	hot	
low	0.1	0.4	0.1	0.6
high	0.2	0.1	0.1	0.4
	0.3	0.5	0.2	1.0

- $H(T) = H(0.3, 0.5, 0.2) = 1.48548$
- $H(M) = H(0.6, 0.4) = 0.970951$
- $H(T) + H(M) = 2.456431$
- **Joint Entropy:** consider the space of (t, m) events $H(T, M) = \sum_{t,m} P(T=t, M=m) \cdot \log \frac{1}{P(T=t, M=m)}$
 $H(0.1, 0.4, 0.1, 0.2, 0.1, 0.1) = 2.32193$

Notice that $H(T, M) \leq H(T) + H(M)$!!!

$$H(T, M) = H(T|M) + H(M) = H(M|T) + H(T)$$

Conditional Entropy

$$H(Y|X) = \sum_{x \in X} p(x) H(Y|X = x) = \sum_{x \in X, y \in Y} p(x, y) \log \frac{p(x)}{p(x, y)}$$

$$P(T = t|M = m)$$

	cold	mild	hot	
low	1/6	4/6	1/6	1.0
high	2/4	1/4	1/4	1.0

Conditional Entropy:

- $H(T|M = low) = H(1/6, 4/6, 1/6) = 1.25163$
- $H(T|M = high) = H(2/4, 1/4, 1/4) = 1.5$
- **Average Conditional Entropy** (aka equivocation):
 $H(T/M) = \sum_m P(M = m) \cdot H(T|M = m) =$
 $0.6 \cdot H(T|M = low) + 0.4 \cdot H(T|M = high) = 1.350978$

Conditional Entropy

$$P(M = m|T = t)$$

	cold	mild	hot
low	1/3	4/5	1/2
high	2/3	1/5	1/2
	1.0	1.0	1.0

Conditional Entropy:

- $H(M|T = cold) = H(1/3, 2/3) = 0.918296$
- $H(M|T = mild) = H(4/5, 1/5) = 0.721928$
- $H(M|T = hot) = H(1/2, 1/2) = 1.0$
- Average Conditional Entropy (aka Equivocation):
 $H(M/T) = \sum_t P(T = t) \cdot H(M|T = t) =$
 $0.3 \cdot H(M|T = cold) + 0.5 \cdot H(M|T = mild) + 0.2 \cdot H(M|T = hot) = 0.8364528$

Conditional Entropy

- Conditional entropy $H(Y|X)$ of a random variable Y given X_i

Discrete random variables:

$$H(Y|X) = \sum_{x \in X} p(x_i) H(Y|X = x_i) = \sum_{x \in X, y \in Y} p(x_i, y_i) \log \frac{p(x_i)}{p(x_i, y_i)}$$

Mixed setting: Continuous (over x) and discrete (over y):

$$H(Y|X) = - \int \left(\sum_{k=1}^K p(y = k|x_i) \log_2(y = k|x_i) \right) p(x_i) dx_i$$

Mutual Information

- Mutual information: quantify the reduction in uncertainty in Y after seeing feature X_i

$$I(X_i, Y) = H(Y) - H(Y|X_i)$$

- The more the reduction in entropy, the more informative a feature.
- Mutual information is symmetric
 - $I(X_i, Y) = I(Y, X_i) = H(X_i) - H(X_i|Y)$
 - $I(Y, X) = \int \sum_k^K p(x_i, y = k) \log_2 \frac{p(x_i, y=k)}{p(x_i)p(y=k)} dx_i$
 - $= \int \sum_k^K p(x_i|y = k)p(y = k) \log_2 \frac{p(x_i|y = k)}{p(x_i)} dx_i$

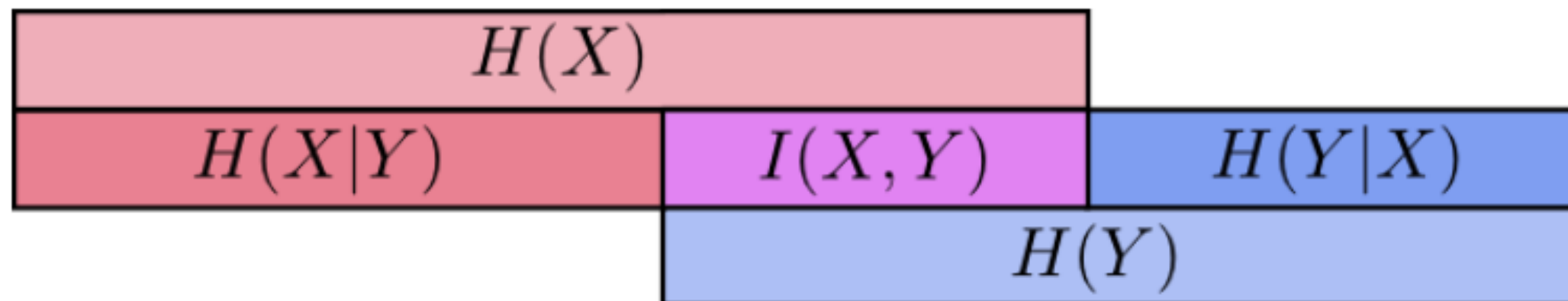
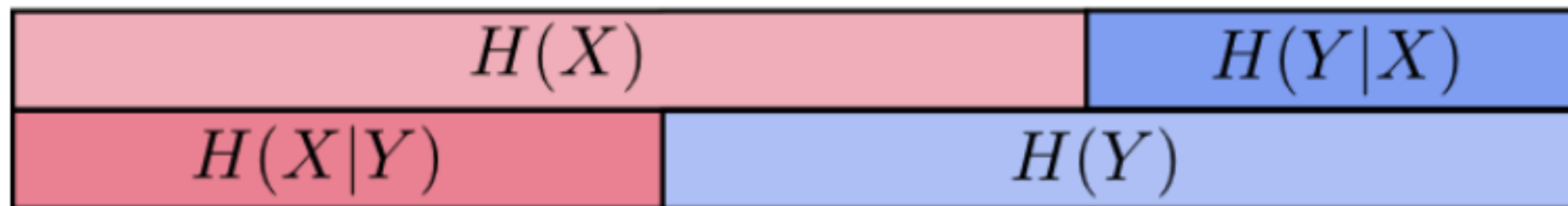
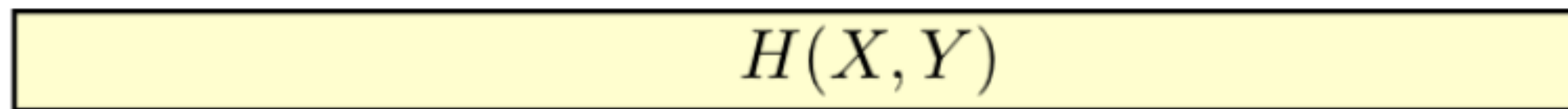
Properties of Mutual Information

$$\begin{aligned} I(X, Y) &= H(X) - H(X|Y) \\ &= \sum_x P(x) \cdot \log \frac{1}{P(x)} - \sum_{x,y} P(x, y) \cdot \log \frac{1}{P(x|y)} \\ &= \sum_{x,y} P(x, y) \cdot \log \frac{P(x|y)}{P(x)} \\ &= \sum_{x,y} P(x, y) \cdot \log \frac{P(x, y)}{P(x)P(y)} \end{aligned}$$

Properties of Average Mutual Information:

- Symmetric
- Non-negative
- Zero iff X, Y independent

CE and MI: Visual Illustration



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- Cross-Entropy and KL-Divergence ←



We will cover this next class in our Optimization lecture

Cross Entropy

Suppose the **true distribution** of data is $P(x)$.

But you assume or model the data using another distribution $Q(x)$.

Cross entropy measures the *average number of bits required* to encode samples from P , **if you use a code optimized for Q instead of P .**

👉 In short: *How inefficient is your code when you use the wrong distribution?*

Entropy Summary

- Entropy ($H(X)$) is the average number of bits needed to encode the random variable X when using an optimal code matched to its distribution $p(x)$.
- Joint Entropy ($H(X,Y)$) is the average number of bits needed to encode both X and Y together, if you use an optimal code matched to their joint distribution.
- Conditional Entropy ($H(Y|X)$) is the average number of extra bits needed to encode Y , once you already know X .
- Cross Entropy is the average number of bits needed to encode data that really follows distribution P , if you instead use a code optimized for another distribution Q .

Cross Entropy

Cross Entropy: The expected number of bits when a wrong distribution Q is assumed while the data actually follows a distribution P

$$H(p, q) = - \sum_{x \in \mathcal{X}} p(x) \log q(x) = H(P) + KL[P][Q]$$

$H(P)$: the true entropy (best-case cost, if you used the right distribution).

$KL[P][Q]$: the extra penalty (inefficiency) you pay for using the wrong distribution Q .

Cross Entropy

Cross Entropy: The expected number of bits when a wrong distribution Q is assumed while the data actually follows a distribution P

$$H(p, q) = - \sum_{x \in \mathcal{X}} p(x) \log q(x) = H(P) + KL[P][Q]$$

This is because:

$$H(p, q) = \mathbb{E}_p[l_i] = \mathbb{E}_p \left[\log \frac{1}{q(x_i)} \right]$$

$$H(p, q) = \sum_{x_i} p(x_i) \log \frac{1}{q(x_i)}$$

$$H(p, q) = - \sum_x p(x) \log q(x).$$

Kullback-Leibler Divergence

Another useful information theoretic quantity measures the difference between two distributions.

$$\begin{aligned}\mathbf{KL}[P(S)||Q(S)] &= \sum_s P(s) \log \frac{P(s)}{Q(s)} \\ &= \underbrace{\sum_s P(s) \log \frac{1}{Q(s)}}_{\text{cross entropy}} - \mathbf{H}[P] = H(P, Q) - H(P)\end{aligned}$$

Excess cost in bits paid by encoding according to Q instead of P .

KL Divergence is
a **KIND OF**
distance
measurement

$$-\mathbf{KL}[P||Q] = \sum_s P(s) \log \frac{Q(s)}{P(s)}$$

log function is
concave or
convex?

$$\begin{aligned}\sum_s P(s) \log \frac{Q(s)}{P(s)} &\leq \log \sum_s P(s) \frac{Q(s)}{P(s)} && \text{By Jensen Inequality} \\ &= \log \sum_s Q(s) = \log 1 = 0\end{aligned}$$

So $\mathbf{KL}[P||Q] \geq 0$. Equality iff $P = Q$

When $P = Q$, $KL[P||Q] = 0$

Take-Home Messages

- Entropy
 - A measure for uncertainty
 - High uncertainty maps to High information and High entropy
 - Why it is defined in this way (optimal coding)
 - Its properties
- Joint Entropy, Conditional Entropy, Mutual Information
 - The physical intuitions behind their definitions
 - The relationships between them
- Cross Entropy, KL Divergence
 - The physical intuitions behind them
 - The relationships between entropy, cross-entropy, and KL divergence